User's Manual for WEB2:

A Stochastic Dynamic Programming Model for Offshore Oil and Gas Leases

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Overview of WEB2

Conceptual Basis

The WEB2 (the acronym for "when exploration begins," version 2) model was developed to determine the optimal time to explore and develop offshore oil and gas leases. Within WEB2, optimal decisions are those that maximize the expected after-tax net present value (ATNPV) of a lease.[1] When the model is run it determines a set of lease operation decision rules that, if followed, will maximize ATNPV. These decision rules show when a lease should be explored and/or developed as a function of: the existing and expected future resource price; the amount of reserves, if any, discovered from exploration; and the number of years remaining in the primary lease term. The model also calculates the expected ATNPV and the expected net economic value (NEV) of a lease if these optimal decision rules are followed.

When a lessee purchases an oil and gas lease from the Government, he has between 5 and 10 years to decide if and when to begin exploring and developing the tract. If the lessee does not begin the exploration/development process within the primary term of the lease, it must be relinquished. The existing models used by the Department of the Interior (DOI) to determine the economic value of a lease assume that exploration always occurs within the primary term of the lease. Usually the exploration date is exogenously specified as an input to the model. This, in turn, sets the development date because development occurs in a fixed number of years after exploration.

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[1] The general structure of WEB2 was patterned after an oil/gas dynamic program entitled "ED&P Optimizer-Social Value Version," originally written by Thomas Teisberg in 1986. The Economic Studies Branch of the Minerals Management Service in Reston, Virginia has a copy of this program.

In WEB2, the timing of exploration and development are determined separately and endogenously. The timing of these activities is based upon the terms of the lease (including royalty rate, rentals, and lease acquisition costs), economic conditions (including the inflation, tax, and discount rates), the geologic potential of the tract, exploration/ production costs, the amount of resource discovered if exploration occurs, the actual level of oil prices when decisions are made, and the expected future price levels. WEB2 also considers the options of not exploring and/or not developing the lease. By using a dynamic programming methodology, in combination with statistical information about the probability of different prices actually occurring in each year, WEB2 determines the lessee's optimal exploration and development strategy. An optimal strategy is identified for each possible price modeled for every year of the primary lease term. Because WEB2 identifies the optimal lease operation strategy, ceteris paribus, the value of a lease as computed by WEB2 will always be greater than or equal to the value estimated by a model that does not optimize exploration and development decisions.

In WEB2, once each year during the primary lease term the lessee has to decide which lease-operation decision is most profitable given the particular circumstances of his lease. The decision options available to the lessee depend on decisions he made in previous years and the number of years remaining in the lease term. In a stochastic dynamic programming model like WEB2, the options available to the decision maker are determined by the value of the model's "state" variables. WEB2 has three state variables: price; time; and lease status.

Table 1 shows the different lease-operation decisions available to the lessee as a function of lease status and time. (The value of the price state variable has no effect on the yearly decision options, but it does affect their profitability.) For example, Table 1 shows that if a lease is unexplored and it is before the last year of the lease's exploration primary term[2] and there is not a binding constraint on exploration, then the lessee can either explore the lease, wait and do nothing for 1 year, or abandon the lease. On the last year of the exploration primary term (i.e., time=EPT) the lessee must either explore the lease or abandon it.

Table 1
Decision Options as a Function
of Lease Status and Time*

Lease Status	Time**	Options***
Unexplored	<EPT and ≥EC	E, W, A
Unexplored	<EPT and <EC	W, A
Unexplored	=EPT and ≥EC	E, A
Unexplored	=EPT and <EC	A
Unexplored	>EPT	N
Explored, oil	<DPT and ≥DC	D, W, A
Explored, oil	<DPT and <DC	W, A
Explored, oil	(=DPT or DPT+1) and ≥DC	D, A
Explored, oil	(=DPT or DPT+1) and < DC	A
Explored, dry	≤EPT+1	A
Explored, dry	>EPT+1	N
Explored	>DPT+1	N
Abandoned		N
Developed		N

* The level of prices has no effect on the decision options available, but it does affect their profitability.

** Measured in years since lease issuance date. When the lease is issued, time=0.

*** EPT=exploration primary term; EC=exploration constraint; DPT=development primary term; DC=development constraint; E=explore; W=wait; A=abandon; N=no action; and D=develop.

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[2] WEB2 has separate lease terms for exploration and development. Normally these two lease terms are set to the same value. However, by setting the development term higher than the exploration term, WEB2 can simulate the effects of relaxing the production diligence requirement.

Table 1 also shows that WEB2 allows the user to impose constraints regarding the first year in which exploration and/or development can occur. For example, if the exploration constraint is set equal to EC=2, then the first year WEB2 will consider exploring the lease is at the beginning of the second year.[3] Normally, the exploration and development constraints should be set equal to zero.

For explored leases, Table 1 shows that the lessee is allowed one extra year beyond the end of development primary term for development decisions. This occurs because in a normal 5-year lease the lessee could begin exploration on the last day of the lease term (i.e., the beginning of year 5 in WEB2). If the lessee were to remain in continuous operation, he would then have the option of deciding whether to develop the lease at the next decision point, the beginning of year 6. In order to allow for this phenomena, WEB2 assumes there will be an extra year for development decisions for any lease that is explored at the end of the exploration primary term. Leases explored prior to the end of the exploration primary term do not get this extra year.

In order to calculate the optimal management strategy for a lease, WEB2 begins at the end of the lease term (plus 1 year for leases explored at the end of the primary term) and determines the optimal decisions for all possible combinations of lease status and price. WEB2 then moves back in time 1 year (i.e., towards the lease issuance date) and re-computes the optimal decisions. When determining the optimal decisions in the penultimate period, WEB2 accounts for how those decisions affect the value of the lease, through the influence of the lease status state variable, in the final period. It also considers the probabilistic nature of price movements from one year to the next and the uncertainties regarding how much resource will be found if the tract is explored. Based

[3] In WEB2 jargon, a lease is issued at the beginning of year 0. Thus, a 5-year lease ends at the beginning of year 5.

on these probabilities and the temporal interdependence in the lease status state variable, WEB2 identifies the optimal decisions for all combinations of price and lease status in the penultimate period. It then moves backward one more period and keeps repeating the process until it identifies the optimal decision on the lease issuance date. This backward-seeking optimization is characteristic of all dynamic programming models.[4]

WEB2 optimizes the after-tax net present value of the lese (ATNPV), but it simultaneously tracks the social economic value of the lease, henceforth referred to as net economic value (NEV). The solution to a WEB2 run shows the optimal decision on the lease issuance data and the ATNPV and NEV associated with that decision. The full solution also shows the optimal decisions and associated values for every year and possible value of the lease status state variable.

A brief mathematical presentation of WEB2 is contained in Appendix C.

Features

WEB2 has a number of significant features, including the following.

- The user can specify the degree of resolution used by WEB2 to model price and reserve uncertainty. More resolution will increase the accuracy of the answer, but it will also increase execution time.

- WEB2 can perform a stochastic delay cost analysis. This analysis tells the user the expected economic value of the

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[4] For a good description of how dynamic programming works see: Hillier, F.S., and Leiberman, G.J. 1974. Operations Research: Second Edition. Chapter 6, San Francisco: Holden-Day, Inc.

lease, discounted to the present, if it is issued at different points in time in the future. The future time points analyzed are specified by the user.

- WEB2 can run in batch mode. This allows the user to successively analyze an unlimited number of prospects in an unattended mode. The results are written to an ASCII file that can be subsequently imported into a spreadsheet program for further analysis.

- WEB2 allows the user to specify different lease terms for exploration and production.

- Separate discount rates can be specified for computing ATNPV and NEV.

- WEB2 optimizes the size of the installed capacity of a platform when it decides to develop a lease. The range of platform sizes considered by WEB2 is specified by the user.

- The development sub-program in WEB2 is calibrated to the existing MMS TSL80 model.

System Requirements

WEB2 will run on any IBM compatible personal computer running under IBM-or MS-DOS. The program itself requires about 90K to load, with additional memory being required for array space. The amount of memory needed for arrays depends on the size of the problem, but normally should be less than 200K. Thus, the program will run on a machine with 384K. A math co-processor is not required, but it will substantially shorten the execution time if it is present.

Uncertainty

Price

One of the key set of inputs for WEB2 are those related to price uncertainty. In WEB2, the real price in the next time period is determined by the following equation:

$$P_{t+1} = P_t(1+g)^{\delta}[T_{t+1}/P_t]^{\alpha}\epsilon_{t+1} \tag{1}$$

where: P_t = real price in period t; g = real trend growth parameter (i.e., the "real trend in oil prices" parameter described in the Inputs section); δ = drift parameter; T_{t+1} = the real trend price in period t+1; and ϵ_{t+1} = a normally distributed error term with mean 0 and standard deviation β. In equation (1) there are an infinite number of possible prices in period t+1 because the error term is a continuous random variable. WEB2 transforms this continuous random variable into a discrete random variable (see equation (2c) for a mathematical statement of the process used) with x possible values, where: x = the number of price grid levels specified by the user.

Equation (1) encompasses two different price models in one equation. First, if δ=0, then the (1+g) term has no direct effect on prices.[5] In this case the next period price is determined by the ratio of the trend price (lagged forward by 1 year) to the actual price, P_t, and the error term. The second model occurs when

[5] There is an indirect effect because g partially determines the level of the trend price. The trend price in period t, T_t, equals $S(1+g)^t$, where S = the trend starting price.

α=0. In this case the ratio of trend price to actual price has no effect, and the g factor and the error term determine prices. Setting δ=0 makes the price errors resemble white noise, or Brownian motion. This is somewhat unrealistic because there is no tendency for the price to move back towards an underlying equilibrium level, regardless of the absolute price level.

The user can set δ and α to any values desired (see input sections for valid ranges), but normally the δ parameter can be set at δ=0. The $(1+g)^{\delta}$ portion of equation (1) was added solely to make WEB2's price process model with TSL80's. The correct values for all of the parameters in the price process model are those that produce a price scenario that is consistent with the user's price expectations.

By changing the values of the trend price, α, or the standard deviation of the error term, β, the user can change the shape of the price process. Two special cases merit mention (both cases assume δ=0). First, if α=1 β=0, then the price path will exactly follow the trend price. This corresponds to the case of perfect price certainty. Second, if α=0, then prices will move randomly, with no tendency to move up or down. The user should experiment with intermediate values of the price parameters to find a set that reflects his expectation of future prices.

To help the user set the values for α and β, equation (1) was fit to historical data on oil prices using maximum likelihood estimation techniques (estimation assumes $\delta=0$). Two data sets were fit: 1949 to 1987 and 1974 to 1987. The coefficients for these parameters are shown in Table 2.[6] The values in the table are probably more helpful for the β parameter than the α parameter.

Table 2

Results of Maximum Likelihood Estimation*

Data Set	α	SE	β	SE
49-87	0.128	0.079	0.164	0.014
74-87	0.204	0.335	0.197	0.056

* A 90 percent confidence interval can be constructed by multiplying the parameter estimated by $\pm$ 1.65 standard errors.

Reserves

The amount of oil contained in a prospect is a random variable until after exploration is completed. WEB2 breaks the reserve uncertainty into two parts: one reflecting the probability of finding nothing and another showing the amounts which may be discovered if the prospect contains hydrocarbons.

[6] To fit the models it was also necessary to estimate the trend price for the first year of the data set (in 1987 dollars) and the real yearly percentage growth rate in the trend. For the 49 to 87 data set, those parameters equaled 6.76 (SE=7.37) and 5.7 (SE=3.6). For the 74 to 87 data set the values were 49.67 (SE=95.84) and -5.4% (SE=21.7).

The probability of finding oil is controlled by the "probability of finding oil" input parameter. One minus the value of this input is the probability of finding no oil if an exploration well is drilled. If oil is discovered, then the number of possible amounts considered by WEB2 is controlled by the "number of reserve grids" input parameter. Each amount considered has an equal probability of occurring.

The actual amounts of oil considered are controlled by the "expected field size" and "standard deviation of field size" inputs. The field size distribution is assumed log-normal and the discrete field sizes selected by WEB2 are based on that underlying continuous distribution.

Specifically, if the mean and standard deviation are μ and σ, respectively, and there are D reserve grids, then the dth (d=1..D) reserve size will equal

$$\text{dth reserve size} = ke^{b+\text{error}(d)} \tag{2}$$

where:

$$b = \ln(\mu) - \tfrac{1}{2}a^2 \tag{2a}$$

$$a = [\ln(\mu^2+\sigma^2) - \ln(\mu^2)]^{\frac{1}{2}} \tag{2b}$$

$$\text{error}(d) = \Phi[(2d-1)/2D]a \qquad (2c)^{7}$$

[7] This equation is also used to generate error terms for different price grids.

Φ = inverse cumulative standard normal function (2d)

$$k = \mu / [\sum_{d=1}^{D} e^{b+error(d)} / D] \qquad (2e)$$

The probability of finding the dth reserve size is (1/D)f, where: f= probability of finding oil. The k term in (2) assures that the expected value of the field sizes used by WEB2 exactly equals the expected field size specified by the user.

Development and Operating Costs

The development costs used by WEB2 are based on the existing MMS TSL80 model. The total cost of developing a platform is

$$\text{Platform Cost} = p(hp-BK+CK)+DK \qquad (3)$$

where: p = peak production capacity of platform measured in barrels per year

$h = \alpha(Q/REFFIELD)^{(\beta+bk-1)}\ FT^{(1-\beta)}$

α,REFFIELD,β,BK,FT,CK DK = TSL80 inputs that are also WEB2 inputs (note: α and β in (3) are not the same as in (1))

Q = fieldsize in barrels

The installed peak production capacity selected depends on the size of the field, the decline rate, and the maximum years at peak production. Given the values of these inputs, the capacity selected by WEB2 will exactly exhaust the resource if production is maintained for the maximum number of years possible.[8] However, in each year of production during the decline phase, WEB2 checks to make sure the lessee is still earning positive profits. If profits turn negative prior to the end of the production profile, WEB2 stops production.

In addition to the platform construction costs, there are three other costs: fixed costs, operating costs, and transportation costs. Fixed costs are borne each year any production occurs and are proportional to peak production capacity (i.e., dollars per barrel of capacity). Operating costs and transportation costs are levied on a per barrel of production basis.

Using WEB2 and Description of Inputs

WEB2 consists of two main files: the "WEB2.EXE" file that contains the executable program; and a data inputs file that controls the operation of WEB2. When the program is delivered the data inputs file is named "WEB2.INP", though the user can use any name for the inputs file. However, as a precaution, the user should save a copy of the original WEB2.INP file in case the format of the input file is altered.

[8] The production profile has a linear build to peak production, a level peak production period, and then an exponential decline.

To run WEB2 the user should place both of these files in the same directory, edit the data input file as desired, and then type "WEB2" followed by a carriage return. The program will pause and prompt the user for the input file name. Once a file name is given, the program will finish running. Alternatively, at the DOS prompt the user can enter "WEB2 filename" where: filename = the name of the input file name. If the input file name is specified directly on the command line, the program will not prompt the user for the input file name. Avoiding the pause for the input file name is useful if the user wants to use a DOS batch file to specify multiple, unattended runs of WEB2.

While WEB2 is executing the program will send messages to the screen to tell the user what task it is working on. When the program is finished it will beep, and show a summary output on the screen. A longer, more detailed output will be written to the output file name specified by the user in the inputs file. This output file can be viewed with a full-screen editor or printed. Appendix B contains an annotated output file.

To edit the "WEB2.INP" file please follow these rules.

- Use an editor that can save the file in plain ASCII or DOS text format.

- o Do not use the TAB command to insert spaces in the file, instead use the space bar. The input routine does not recognize the TAB delimiter.

- o String variables are enclosed in double quotes. The double quotes are already included in the input file so all the user has to do is edit the text inside the double quotes.

- o All variables on the same line of the file must be separated by one or more spaces. For two string variables on the same line, there must be at least one space between each set of double quotes. The file is now set up this way.

- o Do not add or delete any lines to the input file.

The "WEB2.INP" file is divided into five major sections. Each section and the parameters it controls are described below. For each input parameter, its type and valid range are described in parentheses immediately following the parameter's name.

Program Control Inputs

Title: (String; 75 characters or less). The first input for WEB2 is on the first line of the input file. In this line insert any title you wish for a given WEB2 run. The title will be printed at the top of the output file.

Year: (Integer; positive). Insert the starting year for the analysis.

Region: (String; must match one of TSL80 regions specified at the bottom of file). This input tells WEB2 which TSL80 region to use for the analysis.

Price Levels: (Integer; 1 or greater). This parameter controls how much resolution WEB2 will use in the analysis of possible prices. The higher the number the more resolution WEB2 will use and the longer the program's execution time. As a rough rule of thumb, doubling the number of price grid levels will double the execution time. The user should experiment with different numbers of price grid levels to find the best tradeoff between resolution and execution time for a given WEB2 problem.

Reserve Grids: (Integer; 1 or greater). This parameter controls the number of different possible reserve sizes used by WEB2. In a given WEB2 run the user will input the conditional (i.e., given the prospect contains hydrocarbons), expected field size and the conditional standard deviation. WEB2 uses these inputs to generate a range of possible field sizes for analysis (see equation (2)). The number of possible field sizes that are analyzed is determined by the number of reserve grids. The average of the possible field sizes WEB2 analyzes will always exactly equal the conditional, expected field size. Thus, by increasing the value of the reserve grid parameter the user can ask the program to analyze more potential field sizes, yet remain assured that the expected value of the field size will remain constant. Like the price grid level parameter, this parameter will also increase execution time fairly dramatically. Doubling the number of the reserve grid levels will almost double the execution time for the program. Again, experimentation is needed to find the right trade-off between reserve grids and execution time.

Years of Delay Analysis: (Integer; 0 or greater). WEB2 can determine the expected value of a lease if that lease were issued at some user specified date in the future. If this input is zero, WEB2 will not perform any delay analysis; if the input is a positive integer, then WEB2 will analyze the expected economic value of issuing a lease up to that many years in the future. Using the delay analysis option greatly increases the execution time for the program. In general, for each year the delay analysis is conducted the base case program (i.e. with no delay analysis) must be re-run. Thus, running the program with a 1-year delay analysis would almost double its execution time compared to a no delay analysis run (however, the price inputs do not have to be re-computed twice).

Yearly Increment: (Integer; 1 or greater and a factor of the "years of delay analysis" input if that input is non-zero). This parameter controls the number of years by which WEB2 increments for each delay analysis requested. For example if the user has requested 6 years of delay analysis, then this input can be either 1, 2, or 3, or 6. If the yearly increment is set to 1 then 6 years of delay analysis will be executed in 1-year increments. If the yearly increment is set to 2 then the program will analyze the effects of issuing the lease at 2 years, 4 years, and 6 years from the base year. By using a high yearly increment, it is possible to analyze long delays in lease issuance without having to analyze all of the intervening years. If the increment is set to a number that is not a factor of the number of years of delay analysis, 4 or 5 for this example, then the program will note the error and stop.

Output Size: (Integer; 1 or 2). This parameter controls the size of the output file and can be set to either: 1 = small output; or 2 = large output. The normal setting of this input is 1. If the input is set to 2; then all the outputs from a small output will be written to the output file plus WEB2 will output the optimal solution for every possible state a lease could be in in every possible year. Additionally, the long option outputs the expected prices for all years during which oil could feasibly be produced as well as a set of matrices showing price transition probabilities. The price transition probabilities show the likelihood of going from one price level in one year to another price level in another year. Appendix B contains an annotated long output.

Batch Mode: (Integer; 1 or 2). This parameter controls whether WEB2 runs in a regular mode (1), or batch mode (2). In regular mode, the program analyzes only one prospect. In batch mode the program analyzes numerous prospects in a given TSL80 region and outputs the prospect-by-prospect summary results to an ASCII file. This ASCII file is formatted so that it can be subsequently imported into Lotus 1-2-3, or a similar spreadsheet program, for further analysis.

Batch File Name: (String; a valid DOS file name). If a batch analysis has been requested (batch mode = 2), then this parameter tells WEB2 the name of the file containing the batch-mode data. The batch-mode data file must be an ASCII file containing only three columns of numbers. Each prospect should begin on a new line and the columns must be separated by one or more blanks. The first column must contain the probability of a given prospect containing hydrocarbons. This probability should be a "fully-risked" number that reflects all of the types of geologic risks involved including prospect risk, basin risk, and area risk. However, economic risk should not be reflected in this probability because WEB2 will decide for itself what prospects are economic and noneconomic. The second column of the batch-mode data file must contain the expected resource size given the prospect contains hydrocarbons, i.e., the conditional resource size. The third column must contain the standard deviation of the conditional resource size. There is no limit to the number of prospects that can be in the batch-mode data file.

File Name for Expected Prices: (String; a valid DOS file name or blank). When WEB2 runs it must initially calculate all the possible prices during any year where oil might be produced and the probability of moving from a given price in one year to a new price in the next year. This price process computation is somewhat time consuming, especially on slower machines. If a file name is specified for this parameter, WEB2 reads this price information from an existing DOS file (assuming the user has already created the file, see next parameter). If the file name is left blank, by having two double quotations marks next to each other with nothing in the middle, then WEB2 will compute the price information instead of reading the information from an existing file. Reading the price information from an existing file saves the program the burden of recomputing all of the expected prices, but it is a dangerous option to use. The danger occurs because changing any one of a number of the other WEB2 parameters will change the structure of the price information. In particular if the user changes the primary term of the lease, the number of years of delay analysis, the starting price of oil, the trend starting price of oil, the trend growth rate, the inflation rate, the standard deviation of oil prices, the drift parameter, the pull-to-trend in oil prices, or the number of price levels, then the possible prices and associated transition probabilities must be re-computed.

File Name to Write Expected Price Data To: (String; a valid DOS file name or blank). If a file name is specified, then WEB2 will write the price information to this file name. If this parameter is left blank, by having two double quotation marks next to each other with nothing in the middle, then the price information will not be written to an output file. Writing the price information to an output file will not effect the program, except to slow it down slightly during the time the data is written. The possible problems emerge when that file is subsequently re-read into WEB2 (see previous parameter).

Output File: (String; a valid DOS file name). WEB2 will write the results to this file name. Once the program is completed the user can look at the results by either printing this file or by using a full-screen editor. On every non-batch WEB2 run, a summary output is written to the screen. This screen output contains a subset of the full results written to the output file.

Lease Term Inputs

Primary Term of Lease: (Integer; 0 or greater). This input tells the number of years in the lease term. The number of years in the primary lease term determines the number of decision points considered by WEB2. For example, there are six decision points for a 5-year lease. When a lessee buys his lease the lessee must immediately decide whether to explore it or not. This first decision point occurs at the start of year 0. The lessee will then have additional decision points at the beginning of the years 1, 2, 3, 4, and year 5. The beginning of year 5 is 5 years from the beginning of year 0. In a 5-year lease, the beginning of year 5 is the last date at which a lessee can decide whether to explore a lease. If the lease is not explored by the start of year 5, then it must be forfeited. If the lessee explores the lease at the start of year 5, he gets one more decision point at the beginning of year 6 regarding whether to develop the lease or not. A lease explored at, or before, the start of year 4 would not get an extra decision point. This additional decision point occurs because if the lessee begins exploration at the end of the primary lease term and remains in continuous operation, then he can continue holding the lease. Thus, for a 5-year lease, the last possible decision point is the beginning of year 6, at which time the lessee has the option of deciding whether to develop a lease that was explored at the beginning of year 5. Setting this parameter to 0 and the exploration diligence term to 0 forces the lessee to explore or abandon a tract immediately and then, if a find is made, develop or abandon the tract at the start of year 1.

Exploration Diligence Term: (Integer; 0 or greater and less than or equal to the primary lease term). The exploration diligence term specifies the last year, measured in the number of years since the lease issuance date, in which the lessee must either begin exploration or forfeit the lease. By setting the exploration diligence term lower than the primary term of the lease, it is possible to simulate a policy of relaxing the production diligence requirement. Setting this parameter to 0 forces the lessee to explore the lease immediately or forfeit the lease.

Royalty Rate: (Real; between 0 and 1, inclusive). This parameter shows the royalty rate on the lease.

Lease Acquisition Costs: (Real; 0 or greater). This parameter shows the amount the lessee paid to acquire the lease. The lease cost is relevant for two lessee decisions. First, if the lease is abandoned the lessee can write off the lease acquisition cost from his taxes. Second, if the lease is developed the lessee can deplete the bonus during the life of production. Both of these effects are considered by WEB2.

Rental: (Real; 0 or greater). This parameter shows the annual rental rate applicable on the lease. This rental rate is fixed throughout the life of the lease and is not incremented for inflation.

First Year Exploration is Possible: (Integer; 0 or greater). This parameter gives the first year, measured in years since the lease issuance date, when exploration is first possible. If this parameter is greater than 0 it imposes a constraint on the lessee making it impossible for him to consider exploring a lease until the beginning of the first year exploration is possible. Normally this parameter should be set to zero to reflect the usual condition of no constraints. If the input is set to a positive integer, then during the years exploration is constrained, the lessee is not charged rentals. (Note: If the user requests a delay analysis then the WEB2 program internally resets this input as needed for the delay analysis. For example, the value of the 5-year lease issued 1 year from now is computed by resetting the lease term to 6 years, but imposing first-year exploration and development constraints equal to 1. The value of a 6-year lease with these 1-year constraints equals the expected value of issuing a 5-year lease 1 year from now.)

First Year Development is Possible: (Integer; 0 or greater). This parameter gives the first year, measured in years since lease issuance, when development is first possible. Rents are not charged on leases when this constraint is binding.

Financial Inputs

Real Private Discount Rate: (Real; normally non-negative). This parameter shows the real private discount rate used by WEB2. Because WEB2 optimizes private profit, but simultaneously tracks social profits, this discount rate drives the trade-off between present and future. To specify a discount rate of 8 percent, enter 0.08.

Real Social Discount Rate: (Real; normally non-negative). This is the real discount rate used for calculating net economic value.

Discount Timing: This parameter controls whether beginning-of-year or end-of-year discounting is used. When this parameter is set equal to 0, all expenses are assumed to occur at the start of the period (i.e., at the instant the decision is made); if the parameter equal 1, expenses occur at the end of a period. Values between 0 and 1 can be used to reflect during-the-year discounting.

Inflation Rate: (Real; normally non-negative). This is the inflation rate used by WEB2. All of the costs and prices in the program, with the exception of the rental rates, are updated for inflation during each year of the analysis. The first year inflation affects the program is in the beginning of year 1.

Starting Price of Oil: (Real). This is the existing price of oil at the start of the analysis, i.e., the beginning of year zero.

Trend Starting Price: (Real). This is the level of the trend price at the start of the analysis. By setting the trend price above or below the starting price one can simulate the effects of the starting price being lower or higher than expected based on long-run economic considerations.

Standard Deviation of Real Oil Prices: (Real; 0 or greater). This parameter shows the standard deviation of real oil prices from one year to the next. Raising or lowering the value of this input simulates more or less uncertainty in future oil prices. The user should try different levels of standard deviations and look at the subsequent price outputs to find a value that seems consistent with his or her expectation of future oil prices.

Real Trend in Oil Prices: (Real). This is 1/100th of the yearly percentage growth in trend prices that is expected during the duration of the analysis. For example, a 2 percent growth rate is specified as 0.02. This parameter equals g in equation (1) and is also used, in combination with the trend starting price, to calculate the yearly trend price (T_t in equation (1), see Footnote #5).

Pull to Trend: (Real; between 0 and 1, inclusive). This parameter determines the strength of the effect which causes a tendency for actual prices to move toward the trend price. If this parameter is set to zero, then there is no tendency for actual prices to fall or rise toward the trend price. Conversely, if this number is set to one, then prices will have the maximum tendency to fall to the trend prices in any given year. By setting the standard deviation equal to zero and the pull to trend parameter to one, the user can simulate a known certain future oil price path. This is parameter α in equation (1).

Drift Factor: This parameter, which is δ in equation (1), controls the strength of the (1+g) term in equation (1).

ACRS Schedule: (Reals). As the program is now configured it writes off tangible development costs over an 8-year period. The percentage write-off in each year is specified by these eight inputs. The write-off begins when production begins. If the tax law is changed, then these inputs can be modified to reflect the new ACRS Schedule. However, if the ACRS Schedule increases to greater than 8 years, then the program will need to be recompiled to reflect the new tax laws.

Tangible Portion of Development Costs: (Real; between 0 and 1, inclusive). This input shows the proportion of development costs that are considered tangible for tax purposes.

Tax Rate Profits: (Real; between 0 and 1, inclusive). This is the percentage tax rate used to compute private profits.

Geological/Production Inputs

Standard Deviation of Field Size: (Real; 0 or greater). This is the standard deviation of the conditional resource size. WEB2 assumes that the form of the distribution is log-normal. The standard deviation should reflect the standard deviation of the actual field size, not the standard deviation of the logarithm of the field sizes.

Expected Field Size: (Real; positive number). This is the expected resource size given the prospect contains hydrocarbons.

Probability of Finding Oil: (Real; greater than 0 and less than or equal to 1; in batch mode a 0 value is permissible). If a prospect is drilled, this parameter shows the probability of finding hydrocarbons. This probability should reflect all of the geologic risks involved, including area, basin, and prospect risk. It should not, however, reflect economic risk because WEB2 calculates that endogenously.

Exponential Decline Rate: (Real; between 0 and 1). This parameter shows the exponential decline rate used in the declining portion of the production profile. In order to be consistent with TSL80, there is a recommended one-to-one relationship between the exponential decline rate and the maximum number of years in decline. To be consistent with TSL80 the decline rate and the maximum years in decline should be set as is shown in Table 3.

Maximum Years in Decline: (Integer; 2 or greater). This is the maximum number of years in the declining portion of the production profile. The production profile will extract the full resource amount if the decline occurs for the maximum number of years. However, once the peak production period has passed, WEB2 checks each year to make sure that the lessee is still earning a profit

Table 3

Exponential Decline Rate (EDR) Versus Maximum Years in Decline (MYD)

MYD	EDR	MYD	EDR
2	0.7968	22	0.0724
3	0.5312	23	0.0693
4	0.3984	24	0.0664
5	0.3187	25	0.0637
6	0.2656	26	0.0613
7	0.2277	27	0.0590
8	0.1992	28	0.0569
9	0.1771	29	0.0550
10	0.1594	30	0.0531
11	0.1449	31	0.0514
12	0.1328	32	0.0498
13	0.1226	33	0.0483
14	0.1138	34	0.0469
15	0.1062	35	0.0455
16	0.0996	36	0.0443
17	0.0937	37	0.0431
18	0.0885	38	0.0419
19	0.0839	39	0.0409
20	0.0797	40	0.0398
21	0.0759		

on the lease. During the first year when profits turn negative the program will terminate production. Normally this parameter's value is matched to the decline rate as is indicated in Table 3.

Years to Peak: (Integer; 1 or greater). This is the number of years of linear production buildup to peak production. For example, if there are 2 years to peak then the first year's production will be 1/3 of peak production, the second year's production will be 2/3 of peak production, and peak production will begin in year 3.

Minimum Years of Peak: (Integer; 1 or greater). WEB2 checks different number of years at peak production and compares them to determine which one is the most profitable to the lessee. The minimum number of years at peak production considered by the program is specified by this parameter.

Maximum Years of Peak Production: (Integer; 1 or greater and greater than or equal to minimum years of peak production). This specifies the maximum number of years at peak production checked by WEB2. Increasing the range between the minimum and maximum number of years at peak production will increase the execution time of the program. This occurs because for every possible combination of time, price, and reserves, WEB2 will determine the most profitable number of years at peak production. Doubling the range between the minimum or maximum will probably increase the execution time by 60-80 percent. To fix the number of years at peak production, the minimum or maximum parameters should be equal. A large number of years at peak production results in a longer flatter production profile with less installed capacity.

Proportion of Development Costs in Years 1-5: (Real; all numbers between 0 and 1, inclusive, and they must sum exactly to 1). The proportion of the total platform cost that occurs in each year is determined by these five inputs. As soon as the proportions sum to 1, the program assumes the platform is competed. Production will then begin in the next year (unless there is a production lag specified in the TSL80 inputs listed below).

TSL80 Inputs

The Number of TSL80 Regions: (Integer; 1 or greater). This input shows the number of TSL80 regions for which there is data. The number listed here must exactly match the number of TSL80 regions on the input file, or else the program will not run. All of the TSL80 inputs are arranged in a matrix format. The following parameters must be specified for each region.

Region: (String). This input shows the name of each TSL80 region. The same region name must appear twice, once for the first matrix of TSL80 parameters and once for the second matrix of TSL80 parameters.

BK, CK, DK, ALPHA, BETA, REFFIELD, FT: (Real; ranges vary for each input, but all are non-negative). These seven parameters are used to determine the cost of building a development platform. Their use is described in equation (3).

Operating Costs: (Real; 0 or greater). This parameter gives the variable costs of producing a barrel of oil equivalent.

Transportation Costs: (Real; 0 or greater). This is the cost of transporting a barrel of oil equivalent from the prospect to the market. This input is important because royalties are charged on the wellhead price of the oil.

Exploration Costs: (Real; 0 or greater). This is the cost of exploring the prospect to determine whether it contains hydrocarbons. Normally this will equal the costs of drilling one exploration well, but it could represent the cost of drilling many wells if the user feels that many wells will be needed to determine whether the prospect contains hydrocarbons.

Production Lag: (Integer; 0 or greater). When this parameter is set to zero, production will begin the first year after the platform is completed. For example if the platform is completed in 1995, then production will begin in 1996. If the user wants to postpone the start of production relative to the platform completion date, then the user should enter the number of years of delay as the value of this parameter. For example, if the production lag parameter is set equal to 1, then, assuming the platform was completed in 1995, production will begin in 1997.

FFAC1: (Real). The fixed cost per year of operating a platform is determined by the following equation.

Fixed costs = FFAC1 + FFAC2 X peak production capacity.[9]

This level of fixed costs is incurred each year that oil is being produced. These fixed costs should be expressed in base year dollars, just as all the other costs are. WEB2 will automatically

---

[9] Measured in barrels per year.

update these costs for inflation during each year that production continues.

FFAC2: (Real; logically 0 or greater). See FFAC1.

RSP1, RSP2: (Not Used). These parameters stand for region specific parameter 1 and region specific parameter 2. Currently, these inputs are not used by the program and the user can enter any numeric value he desires. However, it is important that a value be entered for these inputs in order that the other inputs are read correctly.

APPENDIX A

LISTING OF INPUT FILE

```
"Test Data Set for WEB2"
************** Program Control Inputs ***************
1989       "Current year"
"GOMD"     "Region, choices: ALARC ALD ALS ATLD ATLS GOMD GOMS PACD PACS"
5          "Number of price levels"
5          "Number of reserve grids"
2          "Number of years of delay analysis, 0= no delay analysis"
1          "Yearly increment for delay analysis"
2          "Output size: 1=small 2=full output"
1          "Batch mode: 1=regular analysis (off), 2=batch analysis (on)"
"5test.prn"  "File name for batch inputs: blank = no batch inputs"
""          "File name containing expected prices: blank= compute"
"web2.pri" "File name to write expected price data to: blank = no write"
"web2.res" "File name for output file, a name is required"
****************** LEASE TERM INPUTS ****************
5          "Primary term of lease"
5          "Exploration diligence term"
0.125      "Royalty rate"
0          "Lease acquisition cost"
11520      "Yearly rental rate for lease"
0          "First year exploration is possible, 0= no constraints"
0          "First year development is possible, 0= no constraints"
**************** FINANCIAL INPUTS *****************
0.08       "Real private discount rate"
0.08       "Real social discount rate"
0          "Discount timing: 0 = start of year; 1 = end of year"
0.05       "Inflation rate"
18         "Starting price of oil"
18         "Trend starting price"
.15        "Standard deviation of real oil prices"
.025       "Real trend in oil prices"
.2         "Pull to trend parameter for price model"
0          "Drift factor for prices: 0 = no drift"
.143 .245 .175 .125 .089 .089 .089 .045  "ACRS Schedule"
0.5        "Tangible portion of development cost"
0.34       "Tax rate on profits"
****************** GEOLOGICAL/PRODUCTION INPUTS ****************
100e6      "Standard deviation of field size"
225e6      "Expected field size"
0.33       "Probability of finding oil"
0.1054     "Exponential Decline rate"
15         "Max Years in decline.  See manual for link to decline rate"
2          "Years to peak"
2          "Min years at peak"
2          "Max years at peak"
.5 .3 .2 0 0 "Proportion of development costs in years 1 to 5"
***************** TSL80 INPUTS BY REGION BELOW THIS LINE ********************
9          "Number of TSL80 regions"
"Region"  "BK"  "CK"     "DK"      "ALPHA" "BETA" "REFFIELD" "FT" "OP$" "TRANS"
"ALARC"   0.50   0    400000000    130000  0.85   275000000   1  3.90  7.40
"ALD"     0.50   0    250000000    105000  0.85   150000000   1  3.90  8.10
"ALS"     0.50   0    100000000     65000  0.85    50000000   1  5.70  3.90
"ATLD"    0.50   0    200000000    100000  0.85   105000000   1  0.60  0.70
"ATLS"    0.50   0     40000000     50000  0.85    10000000   1  0.60  0.70
"GOMD"    0.50   0    300000000    100000  0.85   115000000   1  2.00  1.45
"GOMS"    0.50   0     20000000     40000  0.85     7000000   1  2.00  0.50
"PACD"    0.50   0    250000000    105000  0.85    61000000   1  2.00  0.45
"PACS"    0.50   0     20000000     45000  0.85     7000000   1  2.70  0.45
"------------ More TSL80 Inputs and Other Region Specific Inputs -----------"
"Region"  "EXPLCOST"  "PRODLAG" "FFAC1" "FFAC2" "RSP1" "RSP2"
"ALARC"    4000000       0        0       2       1      1
"ALD"      4000000       0        0       2       1      1
"ALS"      4000000       0        0       2       1      1
"ATLD"     4000000       0        0       2       1      1
"ATLS"     4000000       0        0       2       1      1
```

```
"GOMD"    4000000    0    0    2    1    1
"GOMS"    4000000    0    0    2    1    1
"PACD"    4000000    0    0    2    1    1
"PACS"    4000000    0    0    2    1    1
"---- END OF INPUTS ----"
```

APPENDIX B

ANNOTATED LONG OUTPUT FILE

ANNOTATIONS ARE
INSIDE BOXES

TEST DATA SET FOR WEB2
Date of this run 02-22-1990
Time of this run 13:23:17

TITLE, DATE, TIME

-------------- PROGRAM CONTROL INPUTS --------------
WEB2 designed and programed by Donald H. Rosenthal, USDI

1989 Base year for analysis
GOMD Region being analyzed
5 Number of price grids
5 Number of reserve grids
2 Years of delay analysis
1 Yearly increment in delay analysis
2 Output size 1=small, 2=large
1 Batch analysis 1=regular analysis (off), 2 = batch analysis (on)
5TEST.PRN :File name for batch inputs, blank = no file
WEB2.INP :File name containing inputs for this WEB2 run
:File name for price data inputs, blank = no data
WEB2.PRI :File name for saving price data, blank = no save
WEB2.RES :File name for outputs from this WEB2 run

------------ LEASE TERM INPUTS ---------

5 Primary term of lease
5 Exploration diligence term
.125 Royalty
0 Lease acquisition cost
11520 Rental
0 Years from base year exploration 1st possible
0 Years from base year development is 1st possible

THIS PAGE RE-LISTS THE WEB2 INPUTS. THE USER CAN CHECK THEM TO MAKE SURE THE INPUT FILE WAS READ CORRECTLY.

---------- FINANCIAL INPUTS ------------

.08 Real private discount rate
.08 Real social discount rate
0 Discount timing factor
.05 Inflation rate
18 Starting price
18 Trend starting price
.15 Standard deviation of price growth
.025 Real price trend growth rate
.2 Pull to trend parameter for prices
0 Drift in prices factor
ACRS Schedule
0.143 0.245 0.175 0.125 0.089 0.089 0.089 0.045
.5 Proportion of development costs that are tangible
.34 Tax rate on profits
4000000 Exploration cost
1.45 Transportation cost
2 Operating cost per barrel
0 2 Fixed cost factors 1 and 2

------------ GEOLOGICAL/PRODUCTION INPUTS -----------

1E+08 Standard deviation of field size
2.25E+08 Expected field size
.33 Probability of finding oil
.1054 Exponential decline rate
15 Maximum years in decline
2 Years to peak
2 Minimum years at peak
2 Maximum years at peak
0 Yrs. delay after plat. finished that prod. begins. 0=no delay
Yearly proportions of development costs

```
  0.500  0.300  0.200  0.000  0.000
TSL80 Development parameters
bk=  .5 ck=  0 dk=  3E+08 alpha=  100000
beta=  .85 reffield=  1.15E+08  ftract=  1

 *************** END INPUTS, BEGIN RESULTS REPORTING **********

Possible nominal prices during the primary term (+1) of the lease

  1990  1991  1992  1993  1994  1995
 15.67 15.93 16.55 17.42 18.48 19.70
 17.56 18.26 19.22 20.38 21.71 23.19
 18.99 20.13 21.40 22.82 24.38 26.09
 20.54 22.19 23.84 25.55 27.38 29.34
 23.02 25.53 27.83 30.08 32.38 34.79
```

POSSIBLE NOMINAL AND REAL PRICES FOR EACH PRICE GRID. LAST COLUMN IS THE YEAR AFTER THE END OF PRIMARY LEASE TERM.

```
Possible real prices during primary term (+1) of the lease

  1990  1991  1992  1993  1994  1995
 14.92 14.44 14.29 14.33 14.48 14.70
 16.72 16.56 16.61 16.77 17.01 17.31
 18.09 18.26 18.49 18.77 19.10 19.47
 19.57 20.13 20.59 21.02 21.45 21.90
 21.93 23.16 24.04 24.75 25.37 25.96

 Yearly real trend price starting in  1990
 18.45 18.91 19.38 19.87 20.37 20.87 21.40 21.93 22.48 23.04
 23.62 24.21 24.81 25.43 26.07 26.72 27.39 28.07 28.78 29.50
 30.23 30.99 31.76 32.56 33.37 34.21 35.06 35.94 36.84 37.76
 38.70 39.67
```

TREND PRICES

```
Possible  field sizes and probabilities
   Fieldsize      Probability

+1.219E+08        0.066000
+1.683E+08        0.066000
+2.102E+08        0.066000
+2.624E+08        0.066000
+3.622E+08        0.066000
+0.000E+00        0.670000
```

POSSIBLE EXPLORATION OUTCOMES AND PROBABILITIES

```
---------------- SOLUTION --------------------

E= Explore, W= Wait, D= Develop, A= Abandon, N= No action

Optimal decision at start is W
 Private Value =  9.620974E+07  Social Value =  2.408165E+08
```

SOLUTION AT START AND VALUES

```
Solution for an unexplored tract

1990  1991  1992  1993  1994
W     W     W     W     E
W     W     W     W     E
W     W     W     W     E
E     E     E     E     E
E     E     E     E     E
```

SOLUTION FOR UNEXPLORED TRACT KEYED TO PRICE GRID

```
Marginal (top) and cumulative (bottom) probability of exploration

  1990   1991   1992   1993   1994
0.4000 0.1200 0.0800 0.0640 0.3360
0.4000 0.5200 0.6000 0.6640 1.0000

Private value of these decisions, in current dollars, are:
```

PROBABILITY OF TRACT BEING EXPLORED. EXAMPLE: PROBABILITY OF THIS TRACT BEING EXPLORED IN 1993=0.064; PROBABILITY OF IT BEING EXPLORED BY 1993 OR EARLIER, IS 0.664.

```
+1.033E+08 +1.149E+08 +1.274E+08 +1.401E+08 +1.517E+08
+1.058E+08 +1.185E+08 +1.325E+08 +1.480E+08 +1.647E+08
+1.086E+08 +1.222E+08 +1.372E+08 +1.539E+08 +1.746E+08
+1.109E+08 +1.260E+08 +1.426E+08 +1.613E+08 +1.838E+08
+1.169E+08 +1.342E+08 +1.532E+08 +1.743E+08 +1.980E+08
```

PRIVATE VALUE OF OPTIMAL EXPLORATION STRATEGY KEYED TO PRICE GRID, DISCOUNTED TO YEAR OF DECISION

```
--------------- Full Solution ---------------

Y= 1990 Price= 15.67016 Resr.= +1.219E+08 PV= 7.621267E+07 SV= 2.984484E+08 Action=W
Y= 1990 Price= 15.67016 Resr.= +1.683E+08 PV= 1.79811E+08 SV= 5.144559E+08 Action=W
Y= 1990 Price= 15.67016 Resr.= +2.102E+08 PV= 2.759184E+08 SV= 7.14632E+08 Action=W
Y= 1990 Price= 15.67016 Resr.= +2.624E+08 PV= 3.984322E+08 SV= 9.682749E+08 Action=W
Y= 1990 Price= 15.67016 Resr.= +3.622E+08 PV= 6.382246E+08 SV= 1.46153E+09 Action=W
Y= 1990 Price= 15.67016 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1990 Price= 15.67016 Unexplored PV= 1.032667E+08 SV= 2.551816E+08 Action=W
Y= 1990 Price= 15.67016 Abandoned PV= 0 SV= 0 Action=N
Y= 1990 Price= 15.67016 Developed PV= 0 SV= 0 Action=N
Y= 1990 Price= 17.56054 Resr.= +1.219E+08 PV= 8.081178E+07 SV= 3.070469E+08 Action=W
Y= 1990 Price= 17.56054 Resr.= +1.683E+08 PV= 1.867352E+08 SV= 5.28167E+08 Action=W
Y= 1990 Price= 17.56054 Resr.= +2.102E+08 PV= 2.852111E+08 SV= 7.337507E+08 Action=W
Y= 1990 Price= 17.56054 Resr.= +2.624E+08 PV= 4.112163E+08 SV= 9.97774E+08 Action=W
Y= 1990 Price= 17.56054 Resr.= +3.622E+08 PV= 6.598275E+08 SV= 1.51718E+09 Action=W
Y= 1990 Price= 17.56054 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1990 Price= 17.56054 Unexplored PV= 1.058422E+08 SV= 2.623805E+08 Action=W
Y= 1990 Price= 17.56054 Abandoned PV= 0 SV= 0 Action=N
Y= 1990 Price= 17.56054 Developed PV= 0 SV= 0 Action=N
Y= 1990 Price= 18.99357 Resr.= +1.219E+08 PV= 8.425492E+07 SV= 3.139609E+08 Action=W
Y= 1990 Price= 18.99357 Resr.= +1.683E+08 PV= 1.928236E+08 SV= 5.488776E+08 Action=W
Y= 1990 Price= 18.99357 Resr.= +2.102E+08 PV= 2.952108E+08 SV= 7.620719E+08 Action=W
Y= 1990 Price= 18.99357 Resr.= +2.624E+08 PV= 4.260273E+08 SV= 1.034958E+09 Action=W
Y= 1990 Price= 18.99357 Resr.= +3.622E+08 PV= 6.838223E+08 SV= 1.57243E+09 Action=W
Y= 1990 Price= 18.99357 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1990 Price= 18.99357 Unexplored PV= 1.086113E+08 SV= 2.69664E+08 Action=W
Y= 1990 Price= 18.99357 Abandoned PV= 0 SV= 0 Action=N
Y= 1990 Price= 18.99357 Developed PV= 0 SV= 0 Action=N
Y= 1990 Price= 20.54354 Resr.= +1.219E+08 PV= 8.7538E+07 SV= 3.201748E+08 Action=W
Y= 1990 Price= 20.54354 Resr.= +1.683E+08 PV= 1.978839E+08 SV= 5.592327E+08 Action=W
Y= 1990 Price= 20.54354 Resr.= +2.102E+08 PV= 3.020517E+08 SV= 7.763514E+08 Action=W
Y= 1990 Price= 20.54354 Resr.= +2.624E+08 PV= 4.354505E+08 SV= 1.056667E+09 Action=W
Y= 1990 Price= 20.54354 Resr.= +3.622E+08 PV= 7.001864E+08 SV= 1.616738E+09 Action=W
Y= 1990 Price= 20.54354 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1990 Price= 20.54354 Unexplored PV= 1.109482E+08 SV= 2.815248E+08 Action=E
Y= 1990 Price= 20.54354 Abandoned PV= 0 SV= 0 Action=N
Y= 1990 Price= 20.54354 Developed PV= 0 SV= 0 Action=N
Y= 1990 Price= 23.02183 Resr.= +1.219E+08 PV= 9.245722E+07 SV= 3.301736E+08 Action=W
Y= 1990 Price= 23.02183 Resr.= +1.683E+08 PV= 2.068645E+08 SV= 5.936548E+08 Action=W
Y= 1990 Price= 23.02183 Resr.= +2.102E+08 PV= 3.176621E+08 SV= 8.235532E+08 Action=W
Y= 1990 Price= 23.02183 Resr.= +2.624E+08 PV= 4.62353E+08 SV= 1.168144E+09 Action=D
Y= 1990 Price= 23.02183 Resr.= +3.622E+08 PV= 7.540954E+08 SV= 1.77003E+09 Action=D
Y= 1990 Price= 23.02183 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1990 Price= 23.02183 Unexplored PV= 1.168839E+08 SV= 2.966789E+08 Action=E
Y= 1990 Price= 23.02183 Abandoned PV= 0 SV= 0 Action=N
Y= 1990 Price= 23.02183 Developed PV= 0 SV= 0 Action=N
Y= 1991 Price= 15.92521 Resr.= +1.219E+08 PV= 8.228834E+07 SV= 3.306625E+08 Action=W
Y= 1991 Price= 15.92521 Resr.= +1.683E+08 PV= 1.975928E+08 SV= 5.706405E+08 Action=W
Y= 1991 Price= 15.92521 Resr.= +2.102E+08 PV= 3.043795E+08 SV= 7.927101E+08 Action=W
Y= 1991 Price= 15.92521 Resr.= +2.624E+08 PV= 4.401541E+08 SV= 1.071494E+09 Action=W
Y= 1991 Price= 15.92521 Resr.= +3.622E+08 PV= 7.05722E+08 SV= 1.61892E+09 Action=W
Y= 1991 Price= 15.92521 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1991 Price= 15.92521 Unexplored PV= 1.149474E+08 SV= 2.842187E+08 Action=W
Y= 1991 Price= 15.92521 Abandoned PV= 0 SV= 0 Action=N
Y= 1991 Price= 15.92521 Developed PV= 0 SV= 0 Action=N
```

BEGIN LONG OUTPUT OPTION

OPTIMAL ACTION AND VALUES FOR EVERY COMBINATION OF THE YEAR, PRICE, AND LEASE STATUS STATE VARIABLES

MORE LONG OUTPUT

```
Y= 1991 Price= 18.26148 Resr.= +1.219E+08 PV= 8.975025E+07 SV= 3.441833E+08 Action=W
Y= 1991 Price= 18.26148 Resr.= +1.683E+08 PV= 2.081611E+08 SV= 5.89669E+08 Action=W
Y= 1991 Price= 18.26148 Resr.= +2.102E+08 PV= 3.182808E+08 SV= 8.201475E+08 Action=W
Y= 1991 Price= 18.26148 Resr.= +2.624E+08 PV= 4.592107E+08 SV= 1.115663E+09 Action=W
Y= 1991 Price= 18.26148 Resr.= +3.622E+08 PV= 7.354253E+08 SV= 1.683227E+09 Action=W
Y= 1991 Price= 18.26148 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1991 Price= 18.26148 Unexplored PV= 1.185368E+08 SV= 2.928333E+08 Action=W
Y= 1991 Price= 18.26148 Abandoned PV= 0 SV= 0 Action=N
Y= 1991 Price= 18.26148 Developed PV= 0 SV= 0 Action=N
Y= 1991 Price= 20.12827 Resr.= +1.219E+08 PV= 9.55537E+07 SV= 3.560316E+08 Action=W
Y= 1991 Price= 20.12827 Resr.= +1.683E+08 PV= 2.186318E+08 SV= 6.153743E+08 Action=W
Y= 1991 Price= 20.12827 Resr.= +2.102E+08 PV= 3.33081E+08 SV= 8.536852E+08 Action=W
Y= 1991 Price= 20.12827 Resr.= +2.624E+08 PV= 4.794804E+08 SV= 1.159973E+09 Action=W
Y= 1991 Price= 20.12827 Resr.= +3.622E+08 PV= 7.661851E+08 SV= 1.746889E+09 Action=W
Y= 1991 Price= 20.12827 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1991 Price= 20.12827 Unexplored PV= 1.221862E+08 SV= 3.0139E+08 Action=W
Y= 1991 Price= 20.12827 Abandoned PV= 0 SV= 0 Action=N
Y= 1991 Price= 20.12827 Developed PV= 0 SV= 0 Action=N
Y= 1991 Price= 22.19274 Resr.= +1.219E+08 PV= 1.009034E+08 SV= 3.658949E+08 Action=W
Y= 1991 Price= 22.19274 Resr.= +1.683E+08 PV= 2.262845E+08 SV= 6.29354E+08 Action=W
Y= 1991 Price= 22.19274 Resr.= +2.102E+08 PV= 3.431675E+08 SV= 8.736754E+08 Action=W
Y= 1991 Price= 22.19274 Resr.= +2.624E+08 PV= 4.935833E+08 SV= 1.194585E+09 Action=W
Y= 1991 Price= 22.19274 Resr.= +3.622E+08 PV= 7.985068E+08 SV= 1.870145E+09 Action=D
Y= 1991 Price= 22.19274 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1991 Price= 22.19274 Unexplored PV= 1.259612E+08 SV= 3.16422E+08 Action=E
Y= 1991 Price= 22.19274 Abandoned PV= 0 SV= 0 Action=N
Y= 1991 Price= 22.19274 Developed PV= 0 SV= 0 Action=N
Y= 1991 Price= 25.53094 Resr.= +1.219E+08 PV= 1.092728E+08 SV= 3.833859E+08 Action=W
Y= 1991 Price= 25.53094 Resr.= +1.683E+08 PV= 2.426829E+08 SV= 7.07098E+08 Action=D
Y= 1991 Price= 25.53094 Resr.= +2.102E+08 PV= 3.749796E+08 SV= 9.807286E+08 Action=D
Y= 1991 Price= 25.53094 Resr.= +2.624E+08 PV= 5.431894E+08 SV= 1.326498E+09 Action=D
Y= 1991 Price= 25.53094 Resr.= +3.622E+08 PV= 8.714759E+08 SV= 1.996498E+09 Action=D
Y= 1991 Price= 25.53094 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1991 Price= 25.53094 Unexplored PV= 1.342378E+08 SV= 3.341307E+08 Action=E
Y= 1991 Price= 25.53094 Abandoned PV= 0 SV= 0 Action=N
Y= 1991 Price= 25.53094 Developed PV= 0 SV= 0 Action=N
Y= 1992 Price= 16.54687 Resr.= +1.219E+08 PV= 8.65688E+07 SV= 3.625946E+08 Action=W
Y= 1992 Price= 16.54687 Resr.= +1.683E+08 PV= 2.144507E+08 SV= 6.296084E+08 Action=W
Y= 1992 Price= 16.54687 Resr.= +2.102E+08 PV= 3.325507E+08 SV= 8.742104E+08 Action=W
Y= 1992 Price= 16.54687 Resr.= +2.624E+08 PV= 4.827326E+08 SV= 1.183324E+09 Action=W
Y= 1992 Price= 16.54687 Resr.= +3.622E+08 PV= 7.765928E+08 SV= 1.790597E+09 Action=W
Y= 1992 Price= 16.54687 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1992 Price= 16.54687 Unexplored PV= 1.273748E+08 SV= 3.162199E+08 Action=W
Y= 1992 Price= 16.54687 Abandoned PV= 0 SV= 0 Action=N
Y= 1992 Price= 16.54687 Developed PV= 0 SV= 0 Action=N
Y= 1992 Price= 19.22386 Resr.= +1.219E+08 PV= 9.854503E+07 SV= 3.833326E+08 Action=W
Y= 1992 Price= 19.22386 Resr.= +1.683E+08 PV= 2.309828E+08 SV= 6.582354E+08 Action=W
Y= 1992 Price= 19.22386 Resr.= +2.102E+08 PV= 3.54181E+08 SV= 9.164366E+08 Action=W
Y= 1992 Price= 19.22386 Resr.= +2.624E+08 PV= 5.108204E+08 SV= 1.237516E+09 Action=W
Y= 1992 Price= 19.22386 Resr.= +3.622E+08 PV= 8.171172E+08 SV= 1.867788E+09 Action=W
Y= 1992 Price= 19.22386 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1992 Price= 19.22386 Unexplored PV= 1.325169E+08 SV= 3.266519E+08 Action=W
Y= 1992 Price= 19.22386 Abandoned PV= 0 SV= 0 Action=N
Y= 1992 Price= 19.22386 Developed PV= 0 SV= 0 Action=N
Y= 1992 Price= 21.40266 Resr.= +1.219E+08 PV= 1.083665E+08 SV= 4.037399E+08 Action=W
Y= 1992 Price= 21.40266 Resr.= +1.683E+08 PV= 2.460591E+08 SV= 6.88471E+08 Action=W
Y= 1992 Price= 21.40266 Resr.= +2.102E+08 PV= 3.740417E+08 SV= 9.555985E+08 Action=W
Y= 1992 Price= 21.40266 Resr.= +2.624E+08 PV= 5.366985E+08 SV= 1.287883E+09 Action=W
Y= 1992 Price= 21.40266 Resr.= +3.622E+08 PV= 8.545914E+08 SV= 1.939696E+09 Action=W
Y= 1992 Price= 21.40266 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1992 Price= 21.40266 Unexplored PV= 1.371534E+08 SV= 3.362084E+08 Action=W
Y= 1992 Price= 21.40266 Abandoned PV= 0 SV= 0 Action=N
Y= 1992 Price= 21.40266 Developed PV= 0 SV= 0 Action=N
Y= 1992 Price= 23.83579 Resr.= +1.219E+08 PV= 1.169016E+08 SV= 4.185193E+08 Action=W
Y= 1992 Price= 23.83579 Resr.= +1.683E+08 PV= 2.578411E+08 SV= 7.088726E+08 Action=W
```

MORE LONG OUTPUT

```
Y=  1992  Price=  23.83579 Resr.= +2.102E+08 PV= 3.897409E+08 SV= 9.875544E+08 Action=W
Y=  1992  Price=  23.83579 Resr.= +2.624E+08 PV= 5.626959E+08 SV= 1.37957E+09 Action=D
Y=  1992  Price=  23.83579 Resr.= +3.622E+08 PV= 9.044856E+08 SV= 2.07803E+09 Action=D
Y=  1992  Price=  23.83579 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1992  Price=  23.83579 Unexplored PV= 1.425847E+08 SV= 3.546328E+08 Action=E
Y=  1992  Price=  23.83579 Abandoned PV= 0 SV= 0 Action=N
Y=  1992  Price=  23.83579 Developed PV= 0 SV= 0 Action=N
Y=  1992  Price=  27.83356 Resr.= +1.219E+08 PV= 1.314506E+08 SV= 4.505129E+08 Action=W
Y=  1992  Price=  27.83356 Resr.= +1.683E+08 PV= 2.903519E+08 SV= 8.03985E+08 Action=D
Y=  1992  Price=  27.83356 Resr.= +2.102E+08 PV= 4.38098E+08 SV= 1.106595E+09 Action=D
Y=  1992  Price=  27.83356 Resr.= +2.624E+08 PV= 6.257496E+08 SV= 1.488754E+09 Action=D
Y=  1992  Price=  27.83356 Resr.= +3.622E+08 PV= 9.915254E+08 SV= 2.228749E+09 Action=D
Y=  1992  Price=  27.83356 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1992  Price=  27.83356 Unexplored PV= 1.532089E+08 SV= 3.751684E+08 Action=E
Y=  1992  Price=  27.83356 Abandoned PV= 0 SV= 0 Action=N
Y=  1992  Price=  27.83356 Developed PV= 0 SV= 0 Action=N
Y=  1993  Price=  17.41899 Resr.= +1.219E+08 PV= 8.739149E+07 SV= 3.925049E+08 Action=W
Y=  1993  Price=  17.41899 Resr.= +1.683E+08 PV= 2.283064E+08 SV= 6.881935E+08 Action=W
Y=  1993  Price=  17.41899 Resr.= +2.102E+08 PV= 3.585299E+08 SV= 9.59162E+08 Action=W
Y=  1993  Price=  17.41899 Resr.= +2.624E+08 PV= 5.242142E+08 SV= 1.301693E+09 Action=W
Y=  1993  Price=  17.41899 Resr.= +3.622E+08 PV= 8.478194E+08 SV= 1.965703E+09 Action=W
Y=  1993  Price=  17.41899 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1993  Price=  17.41899 Unexplored PV= 1.400976E+08 SV= 3.510539E+08 Action=W
Y=  1993  Price=  17.41899 Abandoned PV= 0 SV= 0 Action=N
Y=  1993  Price=  17.41899 Developed PV= 0 SV= 0 Action=N
Y=  1993  Price=  20.38352 Resr.= +1.219E+08 PV= 1.069026E+08 SV= 4.262904E+08 Action=W
Y=  1993  Price=  20.38352 Resr.= +1.683E+08 PV= 2.552398E+08 SV= 7.348312E+08 Action=W
Y=  1993  Price=  20.38352 Resr.= +2.102E+08 PV= 3.921595E+08 SV= 1.017395E+09 Action=W
Y=  1993  Price=  20.38352 Resr.= +2.624E+08 PV= 5.662047E+08 SV= 1.374403E+09 Action=W
Y=  1993  Price=  20.38352 Resr.= +3.622E+08 PV= 9.057837E+08 SV= 2.066074E+09 Action=W
Y=  1993  Price=  20.38352 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1993  Price=  20.38352 Unexplored PV= 1.480407E+08 SV= 3.648082E+08 Action=W
Y=  1993  Price=  20.38352 Abandoned PV= 0 SV= 0 Action=N
Y=  1993  Price=  20.38352 Developed PV= 0 SV= 0 Action=N
Y=  1993  Price=  22.821 Resr.= +1.219E+08 PV= 1.218111E+08 SV= 4.52106E+08 Action=W
Y=  1993  Price=  22.821 Resr.= +1.683E+08 PV= 2.758196E+08 SV= 7.704674E+08 Action=W
Y=  1993  Price=  22.821 Resr.= +2.102E+08 PV= 4.178559E+08 SV= 1.061891E+09 Action=W
Y=  1993  Price=  22.821 Resr.= +2.624E+08 PV= 5.982898E+08 SV= 1.429962E+09 Action=W
Y=  1993  Price=  22.821 Resr.= +3.622E+08 PV= 9.540821E+08 SV= 2.189502E+09 Action=D
Y=  1993  Price=  22.821 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1993  Price=  22.821 Unexplored PV= 1.539246E+08 SV= 3.749967E+08 Action=W
Y=  1993  Price=  22.821 Abandoned PV= 0 SV= 0 Action=N
Y=  1993  Price=  22.821 Developed PV= 0 SV= 0 Action=N
Y=  1993  Price=  25.55098 Resr.= +1.219E+08 PV= 1.357856E+08 SV= 4.763043E+08 Action=W
Y=  1993  Price=  25.55098 Resr.= +1.683E+08 PV= 2.951102E+08 SV= 8.038709E+08 Action=W
Y=  1993  Price=  25.55098 Resr.= +2.102E+08 PV= 4.475445E+08 SV= 1.140352E+09 Action=D
Y=  1993  Price=  25.55098 Resr.= +2.624E+08 PV= 6.414813E+08 SV= 1.536256E+09 Action=D
Y=  1993  Price=  25.55098 Resr.= +3.622E+08 PV= 1.019628E+09 SV= 2.303002E+09 Action=D
Y=  1993  Price=  25.55098 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1993  Price=  25.55098 Unexplored PV= 1.613103E+08 SV= 3.964491E+08 Action=E
Y=  1993  Price=  25.55098 Abandoned PV= 0 SV= 0 Action=N
Y=  1993  Price=  25.55098 Developed PV= 0 SV= 0 Action=N
Y=  1993  Price=  30.082 Resr.= +1.219E+08 PV= 1.625904E+08 SV= 5.419993E+08 Action=D
Y=  1993  Price=  30.082 Resr.= +1.683E+08 PV= 3.407222E+08 SV= 9.062671E+08 Action=D
Y=  1993  Price=  30.082 Resr.= +2.102E+08 PV= 5.047696E+08 SV= 1.239443E+09 Action=D
Y=  1993  Price=  30.082 Resr.= +2.624E+08 PV= 7.129337E+08 SV= 1.659982E+09 Action=D
Y=  1993  Price=  30.082 Resr.= +3.622E+08 PV= 1.118262E+09 SV= 2.473797E+09 Action=D
Y=  1993  Price=  30.082 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1993  Price=  30.082 Unexplored PV= 1.743298E+08 SV= 4.189936E+08 Action=E
Y=  1993  Price=  30.082 Abandoned PV= 0 SV= 0 Action=N
Y=  1993  Price=  30.082 Developed PV= 0 SV= 0 Action=N
Y=  1994  Price=  18.48044 Resr.= +1.219E+08 PV= 8.184654E+07 SV= 4.152062E+08 Action=D
Y=  1994  Price=  18.48044 Resr.= +1.683E+08 PV= 2.350766E+08 SV= 7.391448E+08 Action=D
Y=  1994  Price=  18.48044 Resr.= +2.102E+08 PV= 3.768251E+08 SV= 1.036163E+09 Action=D
Y=  1994  Price=  18.48044 Resr.= +2.624E+08 PV= 5.57313E+08 SV= 1.411782E+09 Action=D
```

MORE LONG OUTPUT

```
Y=  1994  Price=  18.48044 Resr.= +3.622E+08 PV= 9.101472E+08 SV= 2.140295E+09 Action=D
Y=  1994  Price=  18.48044 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1994  Price=  18.48044 Unexplored PV= 1.517253E+08 SV= 3.857072E+08 Action=E
Y=  1994  Price=  18.48044 Abandoned PV= 0 SV= 0 Action=N
Y=  1994  Price=  18.48044 Developed PV= 0 SV= 0 Action=N
Y=  1994  Price=  21.71145 Resr.= +1.219E+08 PV= 1.13239E+08 SV= 4.695655E+08 Action D
Y=  1994  Price=  21.71145 Resr.= +1.683E+08 PV= 2.784111E+08 SV= 8.141828E+08 Actio =D
Y=  1994  Price=  21.71145 Resr.= +2.102E+08 PV= 4.309335E+08 SV= 1.129858E+09 Actio =D
Y=  1994  Price=  21.71145 Resr.= +2.624E+08 PV= 6.248738E+08 SV= 1.528771E+09 Action=D
Y=  1994  Price=  21.71145 Resr.= +3.622E+08 PV= 1.003409E+09 SV= 2.301788E+09 Action=D
Y=  1994  Price=  21.71145 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1994  Price=  21.71145 Unexplored PV= 1.646615E+08 SV= 4.081075E+08 Action=E
Y=  1994  Price=  21.71145 Abandoned PV= 0 SV= 0 Action=N
Y=  1994  Price=  21.71145 Developed PV= 0 SV= 0 Action=N
Y=  1994  Price=  24.38303 Resr.= +1.219E+08 PV= 1.367741E+08 SV= 5.103188E+08 Action=D
Y=  1994  Price=  24.38303 Resr.= +1.683E+08 PV= 3.108993E+08 SV= 8.704393E+08 Action=D
Y=  1994  Price=  24.38303 Resr.= +2.102E+08 PV= 4.714988E+08 SV= 1.200101E+09 Action=D
Y=  1994  Price=  24.38303 Resr.= +2.624E+08 PV= 6.755245E+08 SV= 1.616478E+09 Action=D
Y=  1994  Price=  24.38303 Resr.= +3.622E+08 PV= 1.073328E+09 SV= 2.422859E+09 Action=D
Y=  1994  Price=  24.38303 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1994  Price=  24.38303 Unexplored PV= 1.745591E+08 SV= 4.252462E+08 Action=E
Y=  1994  Price=  24.38303 Abandoned PV= 0 SV= 0 Action=N
Y=  1994  Price=  24.38303 Developed PV= 0 SV= 0 Action=N
Y=  1994  Price=  27.37842 Resr.= +1.219E+08 PV= 1.61082E+08 SV= 5.524104E+08 Action=D
Y=  1994  Price=  27.37842 Resr.= +1.683E+08 PV= 3.444542E+08 SV= 9.28543E+08 Action=D
Y=  1994  Price=  27.37842 Resr.= +2.102E+08 PV= 5.133963E+08 SV= 1.27265E+09 Action=D
Y=  1994  Price=  27.37842 Resr.= +2.624E+08 PV= 7.278383E+08 SV= 1.707065E+09 Action=D
Y=  1994  Price=  27.37842 Resr.= +3.622E+08 PV= 1.145543E+09 SV= 2.547906E+09 Action=D
Y=  1994  Price=  27.37842 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1994  Price=  27.37842 Unexplored PV= 1.838266E+08 SV= 4.412938E+08 Action=E
Y=  1994  Price=  27.37842 Abandoned PV= 0 SV= 0 Action=N
Y=  1994  Price=  27.37842 Developed PV= 0 SV= 0 Action=N
Y=  1994  Price=  32.37806 Resr.= +1.219E+08 PV= 1.977703E+08 SV= 6.159399E+08 Action=D
Y=  1994  Price=  32.37806 Resr.= +1.683E+08 PV= 3.95099E+08 SV= 1.01624E+09 Action=D
Y=  1994  Price=  32.37806 Resr.= +2.102E+08 PV= 5.766325E+08 SV= 1.38215E+09 Action=D
Y=  1994  Price=  32.37806 Resr.= +2.624E+08 PV= 8.067965E+08 SV= 1.843789E+09 Action=D
Y=  1994  Price=  32.37806 Resr.= +3.622E+08 PV= 1.254537E+09 SV= 2.736642E+09 Action=D
Y=  1994  Price=  32.37806 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1994  Price=  32.37806 Unexplored PV= 1.980231E+08 SV= 4.658765E+08 Action=E
Y=  1994  Price=  32.37806 Abandoned PV= 0 SV= 0 Action=N
Y=  1994  Price=  32.37806 Developed PV= 0 SV= 0 Action=N
Y=  1995  Price=  19.69945 Resr.= +1.219E+08 PV= 1.100271E+08 SV= 4.776777E+08 Action=D
Y=  1995  Price=  19.69945 Resr.= +1.683E+08 PV= 2.800821E+08 SV= 8.336806E+08 Action=D
Y=  1995  Price=  19.69945 Resr.= +2.102E+08 PV= 4.371851E+08 SV= 1.159865E+09 Action=D
Y=  1995  Price=  19.69945 Resr.= +2.624E+08 PV= 6.370197E+08 SV= 1.57214E+09 Action=D
Y=  1995  Price=  19.69945 Resr.= +3.622E+08 PV= 1.027217E+09 SV= 2.371227E+09 Action=D
Y=  1995  Price=  19.69945 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1995  Price=  19.69945 Unexplored PV= 0 SV= 0 Action=N
Y=  1995  Price=  19.69945 Abandoned PV= 0 SV= 0 Action=N
Y=  1995  Price=  19.69945 Developed PV= 0 SV= 0 Action=N
Y=  1995  Price=  23.19395 Resr.= +1.219E+08 PV= 1.441852E+08 SV= 5.36826E+08 Action=D
Y=  1995  Price=  23.19395 Resr.= +1.683E+08 PV= 3.272343E+08 SV= 9.153293E+08 Action=D
Y=  1995  Price=  23.19395 Resr.= +2.102E+08 PV= 4.960603E+08 SV= 1.261814E+09 Action=D
Y=  1995  Price=  23.19395 Resr.= +2.624E+08 PV= 7.105325E+08 SV= 1.699435E+09 Action=D
Y=  1995  Price=  23.19395 Resr.= +3.622E+08 PV= 1.128695E+09 SV= 2.546947E+09 Action=D
Y=  1995  Price=  23.19395 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y=  1995  Price=  23.19395 Unexplored PV= 0 SV= 0 Action=N
Y=  1995  Price=  23.19395 Abandoned PV= 0 SV= 0 Action=N
Y=  1995  Price=  23.19395 Developed PV= 0 SV= 0 Action=N
Y=  1995  Price=  26.0923 Resr.= +1.219E+08 PV= 1.69837E+08 SV= 5.812447E+08 Action=D
Y=  1995  Price=  26.0923 Resr.= +1.683E+08 PV= 3.626444E+08 SV= 9.766456E+08 Action=D
Y=  1995  Price=  26.0923 Resr.= +2.102E+08 PV= 5.402741E+08 SV= 1.338375E+09 Action=D
Y=  1995  Price=  26.0923 Resr.= +2.624E+08 PV= 7.657389E+08 SV= 1.79503E+09 Action=D
Y=  1995  Price=  26.0923 Resr.= +3.622E+08 PV= 1.204902E+09 SV= 2.678908E+09 Action=D
Y=  1995  Price=  26.0923 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
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Y= 1995 Price= 26.0923 Unexplored PV= 0 SV= 0 Action=N
Y= 1995 Price= 26.0923 Abandoned PV= 0 SV= 0 Action=N
Y= 1995 Price= 26.0923 Developed PV= 0 SV= 0 Action=N
Y= 1995 Price= 29.3435 Resr.= +1.219E+08 PV= 1.96313E+08 SV= 6.270906E+08 Action=0
Y= 1995 Price= 29.3435 Resr.= +1.683E+08 PV= 3.991922E+08 SV= 1.039932E+09 Action=0
Y= 1995 Price= 29.3435 Resr.= +2.102E+08 PV= 5.859084E+08 SV= 1.417395E+09 Action=0
Y= 1995 Price= 29.3435 Resr.= +2.624E+08 PV= 8.227187E+08 SV= 1.893697E+09 Action=0
Y= 1995 Price= 29.3435 Resr.= +3.622E+08 PV= 1.283558E+09 SV= 2.815108E+09 Action=0
Y= 1995 Price= 29.3435 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1995 Price= 29.3435 Unexplored PV= 0 SV= 0 Action=N
Y= 1995 Price= 29.3435 Abandoned PV= 0 SV= 0 Action=N
Y= 1995 Price= 29.3435 Developed PV= 0 SV= 0 Action=N
Y= 1995 Price= 34.78676 Resr.= +1.219E+08 PV= 2.363383E+08 SV= 6.963984E+08 Action=0
Y= 1995 Price= 34.78676 Resr.= +1.683E+08 PV= 4.544436E+08 SV= 1.135605E+09 Action=0
Y= 1995 Price= 34.78676 Resr.= +2.102E+08 PV= 6.548964E+08 SV= 1.536855E+09 Action=0
Y= 1995 Price= 34.78676 Resr.= +2.624E+08 PV= 9.088587E+08 SV= 2.042857E+09 Action=0
Y= 1995 Price= 34.78676 Resr.= +3.622E+08 PV= 1.402467E+09 SV= 3.02101E+09 Action=0
Y= 1995 Price= 34.78676 Resr.= +0.000E+00 PV= 0 SV= 0 Action=A
Y= 1995 Price= 34.78676 Unexplored PV= 0 SV= 0 Action=N
Y= 1995 Price= 34.78676 Abandoned PV= 0 SV= 0 Action=N
Y= 1995 Price= 34.78676 Developed PV= 0 SV= 0 Action=N
```

END FULL SOLUTION

BELOW ARE EXPECTED FUTURE NOMINAL PRICES GIVEN ACTUAL PRICE IN EACH PERIOD. FIRST PRICE IN EACH SERIES KEYS TO PREVIOUS LISTING OF POSSIBLE NOMINAL PRICES SEE EXAMPLE ON NEXT PAGE.

```
grid 1 period 1
 15.67 17.40 19.18 21.02 22.94 24.97 27.10 29.37 31.78 34.36 37.10 40.04 43.19 46.57 50. 9 54.07 58.25 62.74
67.56 72.74 78.32 84.32 90.77 97.70
grid 1 period 2
 15.93 17.89 19.89 21.97 24.12 26.37 28.74 31.23 33.88 36.69 39.69 42.88 46.30 49.96 53.88 58.08 62.59 67.43
72.63 78.23 84.23 90.70 97.65105.12
grid 1 period 3
 16.55 18.72 20.93 23.22 25.59 28.05 30.64 33.37 36.25 39.30 42.55 46.01 49.71 53.66 57.90 62.43 67.30 72.52
78.12 84.15 90.62 97.58105.06113.11
grid 1 period 4
 17.42 19.79 22.21 24.71 27.29 29.98 32.79 35.75 38.87 42.17 45.69 49.43 53.42 57.69 62.25 67.14 72.38 78.01
84.05 90.53 97.51105.00113.06121.72
grid 1 period 5
 18.48 21.06 23.69 26.40 29.20 32.11 35.16 38.36 41.73 45.30 49.10 53.13 57.44 62.04 66.95 72.22 77.87 83.93
90.43 97.42104.92112.99121.66130.99
grid 1 period 6
 19.70 22.49 25.34 28.27 31.30 34.45 37.74 41.20 44.84 48.70 52.79 57.14 61.78 66.73 72.03 77.70 83.78 90.31
97.31104.83112.91121.60130.93140.97
grid 2 period 1
 17.56 19.06 20.63 22.28 24.04 25.91 27.93 30.08 32.40 34.88 37.56 40.44 43.53 46.86 50.44 54.29 58.44 62.90
67.70 72.86 78.42 84.40 90.84 97.77
grid 2 period 2
 18.26 19.96 21.72 23.56 25.51 27.58 29.78 32.14 34.67 37.37 40.27 43.39 46.74 50.34 54.20 58.36 62.83 67.64
72.81 78.38 84.37 90.81 97.74105.20
grid 2 period 3
 19.22 21.10 23.04 25.07 27.21 29.47 31.87 34.43 37.17 40.10 43.24 46.61 50.23 54.11 58.28 62.76 67.58 72.76
78.33 84.33 90.78 97.72105.18113.21
grid 2 period 4
 20.38 22.44 24.56 26.78 29.10 31.56 34.17 36.94 39.91 43.07 46.47 50.10 54.00 58.19 62.68 67.51 72.70 78.28
84.29 90.74 97.68105.15113.19121.83
grid 2 period 5
 21.71 23.95 26.26 28.67 31.19 33.85 36.67 39.67 42.87 46.29 49.95 53.87 58.08 62.59 67.43 72.63 78.22 84.23
90.70 97.64105.12113.16121.81131.11
grid 2 period 6
 23.19 25.63 28.13 30.74 33.47 36.34 39.39 42.63 46.09 49.77 53.72 57.94 62.47 67.33 72.55 78.15 84.17 90.64
97.60105.08113.12121.78131.09141.11
grid 3 period 1
 18.99 20.30 21.69 23.19 24.82 26.59 28.51 30.58 32.83 35.25 37.88 40.71 43.77 47.06 50.61 54.44 58.57 63.01
67.79 72.95 78.49 84.46 90.89 97.82
grid 3 period 2
```

```
 20.13 21.57 23.11 24.76 26.55 28.47 30.55 32.81 35.24 37.86 40.70 43.76 47.05 50.61 54.44 58.56 63.00 67.79
72.94 78.49 84.46 90.89 97.81105.26
grid  3 period 3
 21.40 23.00 24.68 26.49 28.43 30.52 32.78 35.22 37.85 40.68 43.74 47.04 50.60 54.43 58.55 63.00 67.78 72.94
78.49 84.46 90.89 97.81105.26113.28
grid  3 period 4
 22.82 24.57 26.41 28.37 30.48 32.75 35.19 37.83 40.67 43.73 47.03 50.59 54.42 58.55 62.99 67.78 72.93 78.48
84.46 90.89 97.81105.26113.28121.91
grid  3 period 5
 24.38 26.29 28.29 30.42 32.71 35.16 37.81 40.65 43.72 47.02 50.58 54.41 58.54 62.99 67.78 72.93 78.48 84.45
90.88 97.81105.26113.28121.91131.20
grid  3 period 6
 26.09 28.16 30.33 32.65 35.12 37.77 40.63 43.70 47.01 50.57 54.40 58.53 62.98 67.77 72.92 78.47 84.45 90.88
97.80105.26113.28121.91131.20141.20
grid  4 period 1
 20.54 21.61 22.80 24.14 25.63 27.28 29.10 31.09 33.26 35.63 38.20 40.98 44.00 47.26 50.79 54.59 58.70 63.12
67.89 73.03 78.56 84.53 90.95 97.86
grid  4 period 2
 22.19 23.33 24.60 26.03 27.63 29.40 31.35 33.48 35.82 38.36 41.13 44.12 47.37 50.88 54.67 58.76 63.18 67.94
73.07 78.60 84.56 90.97 97.88105.33
grid  4 period 3
 23.84 25.06 26.44 27.99 29.71 31.62 33.72 36.02 38.54 41.28 44.25 47.48 50.97 54.75 58.83 63.24 67.99 73.12
78.64 84.59 91.00 97.91105.35113.36
grid  4 period 4
 25.55 26.89 28.39 30.06 31.92 33.98 36.25 38.74 41.45 44.40 47.61 51.08 54.85 58.91 63.31 68.05 73.17 78.68
84.63 91.04 97.94105.37113.38122.00
grid  4 period 5
 27.38 28.84 30.47 32.28 34.30 36.52 38.97 41.65 44.58 47.76 51.21 54.96 59.01 63.39 68.12 73.23 78.74 84.67
91.07 97.97105.40113.40122.02131.29
grid  4 period 6
 29.34 30.93 32.70 34.67 36.85 39.25 41.90 44.79 47.94 51.37 55.09 59.13 63.49 68.21 73.30 78.80 84.73 91.12
98.01105.43113.43122.04131.32141.30
grid  5 period 1
 23.02 23.67 24.53 25.59 26.86 28.32 29.98 31.84 33.90 36.18 38.67 41.39 44.35 47.56 51.04 54.81 58.88 63.28
68.03 73.15 78.67 84.61 91.02 97.93
grid  5 period 2
 25.53 26.09 26.91 27.97 29.26 30.78 32.52 34.48 36.67 39.09 41.75 44.66 47.83 51.27 55.01 59.05 63.43 68.15
73.25 78.76 84.69 91.09 97.99105.41
grid  5 period 3
 27.83 28.37 29.20 30.30 31.66 33.27 35.12 37.21 39.55 42.14 45.00 48.12 51.52 55.22 59.24 63.58 68.29 73.37
78.86 84.78 91.17 98.05105.47113.46
grid  5 period 4
 30.08 30.64 31.51 32.68 34.13 35.85 37.84 40.09 42.60 45.38 48.45 51.80 55.46 59.44 63.76 68.44 73.50 78.97
84.88 91.25 98.12105.53113.51122.11
grid  5 period 5
 32.38 32.98 33.92 35.18 36.74 38.59 40.72 43.14 45.85 48.84 52.14 55.75 59.69 63.98 68.63 73.66 79.11 84.99
91.35 98.21105.60113.58122.17131.42
grid  5 period 6
 34.79 35.45 36.46 37.82 39.51 41.51 43.81 46.41 49.33 52.56 56.11 60.00 64.24 68.85 73.85 79.27 85.14 91.47
98.31105.69113.65122.23131.48141.44
Price transition probabilities

Year  1
0.20000.20000.20000.20000.2000
0.20000.20000.20000.20000.2000
0.20000.20000.20000.20000.2000
0.20000.20000.20000.20000.2000
0.20000.20000.20000.20000.2000
Year  2
0.60000.20000.20000.00000.0000
0.20000.40000.20000.20000.0000
0.20000.20000.20000.20000.2000
0.00000.20000.20000.40000.2000
0.00000.00000.20000.20000.6000
```

EXAMPLE: GRID 5, PERIOD 3. IF ACTUAL PRICE IS $27.83 IN 1992, THEN EXPECTED NOMINAL PRICE IN 1993 IS $28.37, IN 1994 EXPECTATION IS $29.20, ETC.

MORE LONG OUTPUT. FOR EACH YEAR OF THE PRIMARY LEASE TERM THERE IS A PRICE TRANSITION MATRIX. IT SHOWS THE PROBABILITY OF MOVING FROM

(SEE NEXT PAGE)

```
Year 3
0.60000.20000.20000.00000.0000
0.20000.40000.20000.20000.0000
0.20000.20000.20000.20000.2000
0.00000.20000.20000.40000.2000
0.00000.00000.20000.20000.6000
Year 4
0.60000.20000.20000.00000.0000
0.20000.40000.20000.20000.0000
0.20000.20000.20000.20000.2000
0.00000.20000.20000.40000.2000
0.00000.00000.20000.20000.6000
Year 5
0.60000.20000.20000.00000.0000
0.20000.40000.20000.20000.0000
0.20000.20000.20000.20000.2000
0.00000.20000.20000.40000.2000
0.00000.00000.20000.20000.6000
Year 6
0.60000.20000.20000.00000.0000
0.20000.40000.20000.20000.0000
0.20000.20000.20000.20000.2000
0.00000.20000.20000.40000.2000
0.00000.00000.20000.20000.6000
```

ONE PRICE LEVEL IN YEAR X-1 TO ANOTHER PRICE LEVEL IN YEAR X. EXAMPLE: YEAR 5 MATRIX. LET ROWS BE INDEXED BY I, COLUMNS BY J. THEN ENTRY I,J SHOWS PROBABILITY OF PRICE MOVING FROM GRID LEVEL I IN YEAR 4 TO GRID J IN YEAR 5. THUS, PROBABILITY OF PRICE MOVING FROM $20.38 IN 1993 TO $24.38 IN 1994 IS 0.2.

END LONG OUTPUT, RESUME NORMAL OUTPUT

The value of developing an already discovered 2.25E+08 field in 1989 using a starting price of 18 and nominal trend prices thereafter is as follows(1989 dollars).

ATNPV 3.322835E+08 NEV 8.901361E+08
Years at peak production 2

ASSUMING EXPLORATION IS ALREADY COMPLETED AND IS A SUNK COST, THIS SECTION SHOWS THE ECONOMIC VALUE OF STARTING DEVELOPMENT OF THE EXPECTED FIELD SIZE AT THE BEGINNING OF THE LEASE TERM. OPTIMAL YEARS AT PEAK PRODUCTION IS ALSO LISTED.

----------------- Delay Results ---------------------

Present value of lease (1989 dollars) from private and social perspectives as a function of years of delay in lease issuance

Years Delay	$Private	$Social
0	+9.62097E+07	+2.40817E+08
1	+9.78032E+07	+2.42204E+08
2	+9.88049E+07	+2.39657E+08

RESULTS OF STOCHASTIC DELAY ANALYSIS. FIGURES SHOW THE EXPECTED VALUE OF THE LEASE, DISCOUNTED TO THE PRESENT, IF IT WERE ISSUED IN THE FUTURE.

APPENDIX C

MATHEMATICAL DESCRIPTION OF WEB2

WEB2 optimizes the value of a lease by solving the following recursive equation.

$$V_t(P_t, S_t) = \max_{D_t} [f(P_t, S_t, D_t, t)] + E[V_{t+1}(P_{t+1}, S_{t+1}(S_t, D_t))]/(1+r) \qquad (C1)$$

ST: D_t can only take on values identified as permissible in Table 1 in the text

where: P_t = the nominal price at time t

S_t = the lease status state variable (a lease can be unexplored, explored and undeveloped,[10] developed, or abandoned).

D_t = possible decisions (explore, develop, wait, abandon, or not applicable).

r = the private, nominal discount rate

E = expected value operator

V_t = the value of the lease as a function of the state variables at time t

f( ) = the value of the lease at time t as a function of the state variables and the decision made at time t

[10] Within the explored and undeveloped category there are sub-categories to reflect different amounts of resource discovered as a result of exploration.

$S_{t+1}(S_t, D_t)$ = the lease status state variable in period t+1 as a function of the previous period's status and the decision made at that time period.

WEB2 begins by solving equation (C1) for the last time period and then works backward in time (i.e., towards the lease issuance date) to identify the optimal decision on the lease issuance date.

The expected value operator in equation (C1) is necessary because two of the variables in the dynamic program -- price and reserves -- are random. Thus, for some of the decisions the lessee makes in time period t, he cannot know the actual value that will be realized from that decision. In cases of uncertainty, WEB2 maximizes expected economic value.

The Exploration Decision

The private value of exploring a lease, as measured in time t using beginning of year discounting, is[11]

$$V_{et} = -(1-tax)(C_{et}+N) + E_{t+1}[lease|explored]/(1+r) \qquad (C2)$$

where: V_{et} = private value of an exploration decision in

[11] The social value of exploration is

$$-C_{et}+E_{t+1}[NEV|explored]/(1+r')$$

where: r' = nominal social discount rate.

time t

N = annual rental rate on lease

tax = marginal tax rate

C_{et} = nominal exploration cost at t

Equation (C2) shows that there are two components of value to an exploration decision. The first component is the after-tax cost of drilling a well(s), which is equal to $-(1\text{-tax})(C_{et}+N)$. The second component of value is the expected value of the lease in period t+1 given that it was explored in period t. Once a lease is explored its value in the next year will depend on how much resource was found and future prices. From WEB2's perspective, exploration is potentially valuable because it changes the value of the lease status state variable, thereby, creating the option of developing the lease. At the time of the exploration decision, both reserves and future prices are unknown. Thus, the expected value operator in (C2) operates over both prices and reserves.

The Development Decision

The private value of developing an already explored lease, as measured in time t and using beginning of year discounting, is

$$V_{pt} = \sum_{t=a}^{b}[Y_t(1-\text{tax})+\text{tax}(D_t+B_t)-T_t]/(1+r)^{t-a} \qquad \text{(C3)}$$

where: V_{pt} = private value of development in time t, i.e. ATNPV[12]

Y_t = taxable income in time t

= $G_t-O_t-F_t-R_t-I_t$

G_t = gross revenue at time t

= production at t x wellhead price at t

O_t = variable operating costs in t

F_t = fixed operating costs in t

R_t = royalty in t

I_t = intangible development costs in t

D_t = depreciation claimed in t

B_t = bonus depletion in t

T_t = tangible development costs in t

a = year development begins

b = year production ends (in WEB2 this is the lesser of when profits first turn negative or when the resource is exhausted)

There is a key difference between (C2) and (C3): in (C3) the value of a development decision does not depend on the value of the lease status state variable in the next time period; for the exploration decision in (C2), the value of the lease status variable in the next period matters. This occurs because WEB2 assumes that once a lease is developed, all of the benefits of development accrue in the period when the decision to develop the lease is made. The

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[12] The social value of development, NEV, is

$$\sum_{t=a}^{b}[G_t-O_t-F_t-I_t-T_t]/(1+r')^{t-a}$$

where: r' = the nominal social discount rate.

benefits are, as is shown in (C3), the expected ATNPV of development (NEV is also tracted). For modeling purposes, once a lease is developed, it has no value in future time periods. The "N" decisions in Table 1 of the text reflects this feature.

In order to compute ATNPV, WEB2 uses the expected future price path at the time the development decision is made. Each time-period/price-grid combination has a different expected future price path. Using the expected future price path for development decisions, as opposed to considering all possible prices paths, greatly simplifies WEB2 -- in fact, it is a key step that enable WEB2 to run on a microcomputer. In order to consider all of the possible price paths it would have been necessary to add one or more state variables to the model. This would, in turn, markedly increase the computational burdens on the model.

Using a single expected price vector for development decisions is valid if

$$E[D(P)] = D(E[P]) \tag{C4}$$

where: D(P) = the value of development as a function of present and future prices; and E is the expected value operator. If D(P) is linear in prices, then (C4) would be true and WEB2's modeling approach would be exactly correct. However, DCP is not linear in prices because prices affect the abandonment date of a field and may also affect the rate of resource extraction. However, these effects probably do not significantly affect ATNPV, and (C4) is judged approximately correct.[13]

The Wait Decision

The private value of waiting, as measured in time t using beginning of year discounting, is

$$V_{wt} = -R(1-tax) + E_{t+1}[lease|wait\ in\ t]/(1+r) \qquad (C5)$$

where: V_{wt} = value of waiting in time t

R = annual rental amount (rentals are levied only if the exploration and development constraints, i.e., EPT and DPT in Table 1, are not binding).

Waiting is advantageous to the lessee when the expected discounted value of the lease a year later is greater than any of the other decision options available. When a lessee waits, the value of the lease status state variable does not change. The social value of waiting is the same as the private value, except that rentals are not charged and the expected lease value in period t+1 is computed using NEV.

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[13] Because of these effects, E[D(P)] is slightly greater than D(E[P]). Thus, WEB2 tends to somewhat undervalue development. A Monte Carlo model could be used to help estimate the degree of understatement.

The Abandonment Decision

The private value of abandoning an undeveloped lease, as measured in any time period using beginning of year discounting, is

$$V_A = \text{(lease cost)(tax)} \qquad \text{(C6)}$$

Once a lease is abandoned, it has no value in future time periods. Only undeveloped leases can be abandoned. Abandoning a lease has no social value.

The Not Applicable Decision

If a lease has already been abandoned or developed, then it has no value in future time periods. In order to create a place a holder for these cases, the "not applicable" decision was created. A not applicable decision has zero private and zero social value. The not applicable decision is also relevant for decisions about whether to explore a lease after the end of the exploration primary term.